SAVEETHA SCHOOL OF ENGINEERING
C PROGRAMMING ASSIGNMENT – Grama Jyothi Billing Program

Field	Entry
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Register Number	1912260562
Course Code & Name	CSA0209 C PROGRAMMING
Slot	B
Assignment / Task Title	Design, Implement and Validate a Reliable Billing Program
GitHub Repository URL	https://github.com/bavadarini28 /CSA0209-1912260562-BAVADARANI.S.git
Submission Date	12 NOVEMBER 2026

TASK 1 – Understand the Problem and Identify the Data

Aim

To identify every input, output, constant and rule of the billing problem and justify each data item.

Analysis

Item	Kind	Justification
Household number	Input	Identifier, positive, up to 800,000 in a district; never used in arithmetic, so it is a label.
Previous / current reading	Input	Non-negative whole numbers in 0–65535 (16-bit register). Never fractional.
Category	Input	Small code (e.g. domestic = 1); a char or enum is enough.
Units consumed	Computed	current − previous, with rollover correction; whole and non-negative.
Energy charge	Computed	Slab-wise; fractions of a rupee arise (Rs 2.25) so store in paise.
Fixed charge	Constant	Rs 50 = 5000 paise; does not change in a run.
Total	Output	fixed + energy, in paise; district totals reach billions of paise.
Slab limits & rates	Constants	100, 200 units; 0, 225, 450 paise. Must live in one place so a revision is applied everywhere.

Constants: slab limits, rates, fixed charge, register maximum (65535). **Negative:** only a naive current − previous after rollover (e.g. 120 − 65400); a valid bill is never negative. **Fractional:** rates in rupees (2.25), solved by using paise. **Very large:** district totals (800,000 × ~Rs 275 = 22 billion paise, beyond a 32-bit int).

Rules a correct bill must satisfy:

- Bill ≥ fixed charge ≥ 0 (never negative).
- Total = fixed charge + energy charge, exactly, to the paisa.
- Bill is non-decreasing as units increase (monotonic).
- Units = (current − previous) mod 65536.
- Sum of all bills = sum of the parts, regardless of order of addition.

Assumptions: one meter register per house; at most one rollover per month; a month never exceeds 5,000 units (so a "current < previous" jump larger than that is treated as a data error, not a rollover); one tariff for all categories; amounts in Indian rupees.

Alternative Concept – readings stored as text ("01530")

Each reading would be kept as char[6]. To subtract, it must be converted digit by digit (value = value*10 + digit), and to compare it can be compared directly with strcmp when zero-padded to the same width, or after conversion.

	Integer (unsigned short)	**Text (d digits)**
Convert	Once at input: O(d)	O(d) every time it is used
Compare / subtract	O(1)	O(d) (strcmp) or O(d) convert + O(1)
Space per reading	2 bytes	d + 1 bytes (6 bytes for 5 digits)
For 800,000 readings	1.6 MB	4.8 MB

Since $d \leq 5$ is tiny, the time difference is small, but integers are 3× smaller, avoid repeated conversion and remove parsing bugs. Text is only preferable for keeping leading zeros for display.

Result: Every data item, constant and rule is identified and justified.